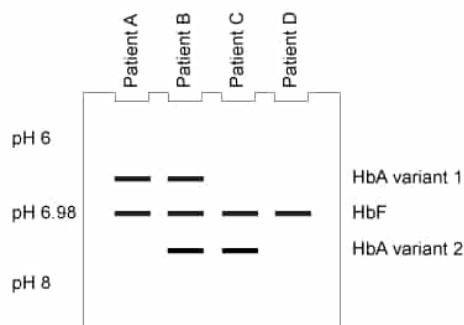


Fetal hemoglobin (HbF) is the predominant form of hemoglobin in newborns, but some adult hemoglobin (HbA) may be expressed as well. The wild-type variant of HbA contains a glutamate (E) residue at position 6. A heterozygous E6V mutation in HbA causes sickle cell trait whereas a homozygous E6V mutation causes sickle cell disease. The isoelectric focusing gel below shows hemoglobin samples from four newborn patients. Which HbA variant most likely has the E6V mutation, and which newborn most likely has sickle cell disease?

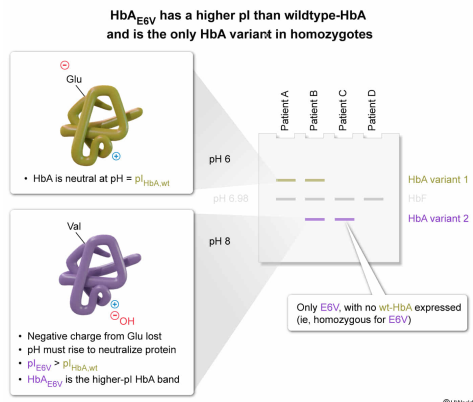


- ☐ A. HbA variant 1 has the E6V mutation and Patient A has sickle cell disease [17%]
- ☐ B. HbA variant 1 has the E6V mutation and Patient B has sickle cell disease [23%]
- ✓ ☒ C. HbA variant 2 has the E6V mutation and Patient C has sickle cell disease [46%]
- ☐ D. HbA variant 2 has the E6V mutation and Patient D has sickle cell disease [12%]

Correct

45% answered correctly

Explanation:



Isoelectric focusing (IEF) is a powerful analytic technique that can detect and distinguish proteins with mutations that affect their **ionizability**. In an IEF experiment, proteins are loaded into a gel that has a pH gradient, and the proteins migrate through the gel until they reach the point where the pH equals the protein's **isoelectric point** (ie, pI value), which is the pH where the protein's overall charge is electrically neutral. Mutations that add, remove, or change an ionizable side chain will alter the charge of the protein at any given pH, and therefore will alter the pI value of the mutant compared to wild-type.

Wild-type HbA has a negatively charged **glutamate** at position 6. When the pH equals the isoelectric point of wild-type HbA, the overall charge of the protein is neutral. The notation **E6V** means that the glutamate (E) at position 6 has been replaced by **valine** (V), which is neutral and nonionizable.

The **loss of the negative** glutamate means that the E6V mutant is **positively** charged when the pH equals the isoelectric point of the wild-type form. To neutralize the E6V mutant, the pH must **increase** (ie, add OH⁻ and remove H⁺) to neutralize a positive charge from the mutant protein; therefore, **mutant HbA has a higher isoelectric point** than wild-type HbA.

The gel shows that *HbA variant 2 has a higher isoelectric point* than variant 1, and therefore **variant 2 most**

likely has the E6V mutation. Patient C has variant 2 only, indicating that this patient is **homozygous** for the E6V mutation. Consequently, **Patient C most likely has sickle cell disease.**

(Choice A) Patient A is homozygous HbA variant 1, which is the wild-type form.

(Choice B) Patient B expresses both HbA variants; patient B is heterozygous and most likely has sickle cell *trait*.

(Choice D) Patient D does not yet express either HbA variant, so no conclusions can be drawn about whether this patient has sickle cell disease.

Educational objective:

Isoelectric focusing gels are used to screen mutations that alter ionizable amino acids based on their effect on a protein's isoelectric point (pI). The removal of a negatively charged amino acid causes an increase in the mutant's pI relative to wild-type.

Time spent:0 sec

Subject:
Biochemistry

QID:403682

System:
Amino Acids and Proteins

The side chain of histidine has a pK_a of approximately 6, allowing it to readily exchange protons with its environment at pH levels near physiological pH. For this reason, histidine is commonly found in the active sites of enzymes, where it can participate in acid-base catalysis.

Some theories of the origin of life propose that dipeptides containing two histidine residues (His-His) provided the catalytic activity required to generate more complex peptides. Indeed, when glycine is incubated with His-His, the rate of polyglycine formation is increased significantly relative to glycine under identical conditions except with His-His absent. Thus, it is hypothesized that His-His formation was a critical step in the eventual formation of proteins.

To test whether His-His could spontaneously form in prebiotic conditions, 4 mg of free histidine was dissolved in 1 mL of aqueous solution and mixed with condensing agents believed to have been present on prebiotic Earth. This was repeated with solutions at different initial pH levels. The solutions were then heated, and the water was allowed to evaporate over the course of 24 hours. The resulting mixtures were each redissolved in phosphate-buffered saline at pH 7 and analyzed. Table 1 summarizes the results of the experiment.

Table 1 Amount of His-His Recovered After Evaporation of Solutions at Various Initial pH Levels Followed by Redissolution

Initial pH	his-his recovered
5	0.16 mg
6	0.36 mg
7	0.52 mg

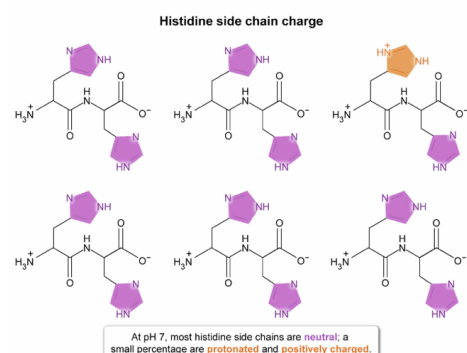
Based on the pK_a of the histidine side chain given in the passage, which of the following best describes the charges of the His-His dipeptide side chains at pH 7?

- ☐ A. The side chains all have a charge of 0. [20%]
- ☐ B. The side chains all have a charge of +1. [18%]
- ✓ ☒ C. Most of the side chains have a charge of 0, but a small percentage have a positive charge. [41%]
- ☐ D. Most of the side chains have a charge of +1, but a small percentage are neutral. [19%]

Correct

41% answered correctly

Explanation:



Seven of the amino acids found in proteins have **ionizable side chains** (ie, side chains that can readily exchange protons with their environment). The charge of an ionizable side chain is determined by its **pK_a value** and the **pH of the solution** in which it is found.

If the **ambient pH is less than the pK_a** of the side chain, the **majority of the side chains** in solution are **fully protonated**. If the **pH is greater than the pK_a** of the side chain, the **majority of the side chains are deprotonated**. The exact percentage of side chains that are protonated and deprotonated at any pH can be calculated using the **Henderson-Hasselbalch equation**.

For some ionizable amino acids, a **fully protonated side chain has a positive charge** and a **deprotonated side chain is neutral** (eg, Arg, Lys, His). For other ionizable amino acids, a **fully protonated side chain is neutral** and a **deprotonated side chain is negatively charged** (eg, Glu, Asp, Tyr, Cys).

In this case, the dipeptide consists of two **histidine** residues, so each side chain can either have a positive charge (protonated) or be neutral (deprotonated). The passage states that the pK_a of the histidine side chain is approximately 6, and the question asks about His-His at **pH 7**. Because the **pH is greater than the pK_a** of each side chain, **most of the side chains in solution are deprotonated, making them electrically neutral**. However, based on the Henderson-Hasselbalch equation, a **small percentage (~9%) of the side chains are protonated and have a positive charge**.

(Choice A) At pH 7, *most* histidine side chains are electrically neutral, but **approximately 9%** are protonated and have a charge of +1.

(Choices B and D) Only a small percentage of histidine side chains have a charge of +1 at pH 7.

Educational objective:

Histidine has an ionizable side chain that can either be positively charged or neutral. At pH 7, the majority of histidine side chains in solution are neutral, but a small percentage have a positive charge.

Time spent:0 sec
Subject:
Biochemistry

QID:403683
System:
Amino Acids and Proteins

The side chain of histidine has a pK_a of approximately 6, allowing it to readily exchange protons with its environment at pH levels near physiological pH. For this reason, histidine is commonly found in the active sites of enzymes, where it can participate in acid-base catalysis.

Some theories of the origin of life propose that dipeptides containing two histidine residues (His-His) provided the catalytic activity required to generate more complex peptides. Indeed, when glycine is incubated with His-His, the rate of polyglycine formation is increased significantly relative to glycine under identical conditions except with His-His absent. Thus, it is hypothesized that His-His formation was a critical step in the eventual formation of proteins.

To test whether His-His could spontaneously form in prebiotic conditions, 4 mg of free histidine was dissolved in 1 mL of aqueous solution and mixed with condensing agents believed to have been present on prebiotic Earth. This was repeated with solutions at different initial pH levels. The solutions were then heated, and the water was allowed to evaporate over the course of 24 hours. The resulting mixtures were each redissolved in phosphate-buffered saline at pH 7 and analyzed. Table 1 summarizes the results of the experiment.

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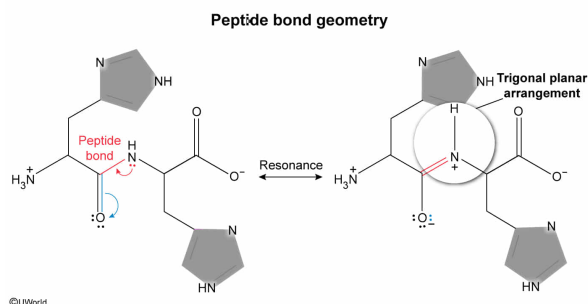
What is the molecular geometry of the nitrogen atom in the peptide bond that links the two histidine residues in His-His?

- ☐ A. Tetrahedral [21%]
- ☐ B. Trigonal pyramidal [34%]
- ☐ C. T-shaped [4%]
- ✓ ☒ D. Trigonal planar [39%]

Correct

38% answered correctly

Explanation:



The **molecular geometry** of an atom refers to the arrangement of the substituents around that atom. Molecular geometry is determined by the number of **electron domains** (ie, regions of electron density) around the central atom, and how many of those electron domains are electron pairs in **sigma bonds** versus how many are **lone electron pairs**.

The nitrogen atom in the bond that links the two histidine residues in His-His is part of a **peptide bond**, which is a type of **amide bond**. The nitrogen atom in a peptide bond is commonly drawn surrounded by four electron pairs: one pair forming a bond to hydrogen, one pair forming a bond to the α -carbon of one amino acid residue, one pair forming a bond to the carbonyl carbon of the other amino acid residue, and one lone electron pair.

However, the lone electron pair on the nitrogen atom participates in **resonance** with the carbonyl carbon, forming a pi bond. This resonance gives the peptide bond **partial double-bond character** and causes the nitrogen atom to have **three electron domains instead of four**. **Double bonds** cause all bonds to the central atom to be in the **same plane**, so the resonance effect in peptide bonds gives the nitrogen a **trigonal planar** geometry.

(Choice A) **Tetrahedral** geometry corresponds to atoms with four electron domains, all of which are bonds (ie, no lone electron pairs). For example, the **α -carbons** of amino acids have tetrahedral geometry.

(Choice B) If each of the four electron pairs formed a distinct domain, the nitrogen atom would have **trigonal pyramidal** geometry. However, the lone electron pair does not form its own domain because it participates in a pi bond through resonance.

(Choice C) **T-shaped** geometry corresponds to an atom with five or six electron domains, three of which are bonds and the rest of which are lone pairs. None of the atoms in His-His have T-shaped geometry.

Educational objective:

Peptide bonds have partial double-bond character due to resonance between the lone pair of the nitrogen atom and the carbonyl carbon. As such, the nitrogen atoms in a peptide bond have trigonal planar geometry.

The side chain of histidine has a pK_a of approximately 6, allowing it to readily exchange protons with its environment at pH levels near physiological pH. For this reason, histidine is commonly found in the active sites of enzymes, where it can participate in acid-base catalysis.

Some theories of the origin of life propose that dipeptides containing two histidine residues (His-His) provided the catalytic activity required to generate more complex peptides. Indeed, when glycine is incubated with His-His, the rate of polyglycine formation is increased significantly relative to glycine under identical conditions except with His-His absent. Thus, it is hypothesized that His-His formation was a critical step in the eventual formation of proteins.

To test whether His-His could spontaneously form in prebiotic conditions, 4 mg of free histidine was dissolved in 1 mL of aqueous solution and mixed with condensing agents believed to have been present on prebiotic Earth. This was repeated with solutions at different initial pH levels. The solutions were then heated, and the water was allowed to evaporate over the course of 24 hours. The resulting mixtures were each redissolved in phosphate-buffered saline at pH 7 and analyzed. Table 1 summarizes the results of the experiment.

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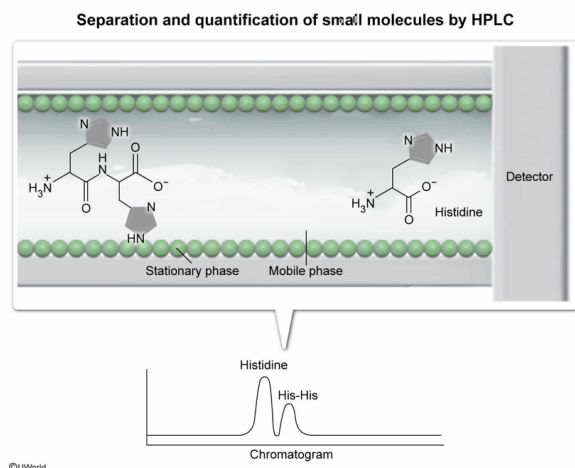
The redissolved mixtures were most likely subjected to which of the following techniques to separate His-His from monomeric histidine and generate the data in Table 1?

- ☐ A. Gel electrophoresis and Coomassie staining [49%]
- ✔ ☒ B. High-performance liquid chromatography [37%]
- ☐ C. Spectrophotometry at 280 nm wavelength [5%]
- ☐ D. Northern blot and densitometry analysis [7%]

Correct

38% answered correctly

Explanation:



To analyze the amounts of different compounds in a solution, the compounds must be **separated** and then **quantified**. A variety of techniques exist to separate molecules by various chemical properties.

High-performance liquid chromatography (HPLC) is a technique in which a mixture of molecules enters a column packed with beads. The beads may be hydrophilic (normal-phase HPLC) or hydrophobic (reverse-phase HPLC). As the mixture moves through the column, **different compounds** in the mixture **experience different strengths of attraction for the beads** (ie, the stationary phase) and for the solvent in which they are dissolved (ie, the mobile phase). Molecules with a **stronger attraction to the beads stay in the column longer** and are separated from molecules with a weaker attraction to the beads.

At the end of the column, a detector measures the **absorbance** of each compound at a particular **wavelength** (usually an ultraviolet wavelength). This result can be compared to standards of a known amount to **determine the amount** of each compound in a sample.

HPLC is a good technique for **separating** small organic molecules such as **individual amino acids and dipeptides** because small changes in chemical structure can result in large changes in affinity for the column. Of the options given, **HPLC is the best for measuring amounts of His-His, as reported in Table 1.**

(Choice A) **Gel electrophoresis** and Coomassie staining are good for separating and quantifying large molecules like proteins, but electrophoresis generally does not have the resolution needed to separate dipeptides (eg, His-His) from individual amino acids (eg, individual histidine molecules), nor would Coomassie staining be sensitive enough to detect dipeptides.

(Choice C) The side chain of histidine absorbs weakly at 280 nm, but simple **spectrophotometry** would not separate His-His from individual histidine residues, so this technique would not reveal how much of the absorbance was due to His-His specifically.

(Choice D) **Northern blots** are used to detect and quantify *RNA* samples, not peptides or amino acids.

Educational objective:

High-performance liquid chromatography (HPLC) is a good technique for separating and quantifying small organic molecules such as amino acids and peptides.

Time spent:0 sec

Subject:
Biochemistry

QID:403685

System:
Amino Acids and Proteins

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Some theories of the origin of life propose that dipeptides containing two histidine residues (His-His) provided the catalytic activity required to generate more complex peptides. Indeed, when glycine is incubated with His-His, the rate of polyglycine formation is increased significantly relative to glycine under identical conditions except with His-His absent. Thus, it is hypothesized that His-His formation was a critical step in the eventual formation of proteins.

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6	0.36 mg
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Which of the following observations would provide additional support for the hypothesis that His-His formation was critical to prebiotic protein formation?

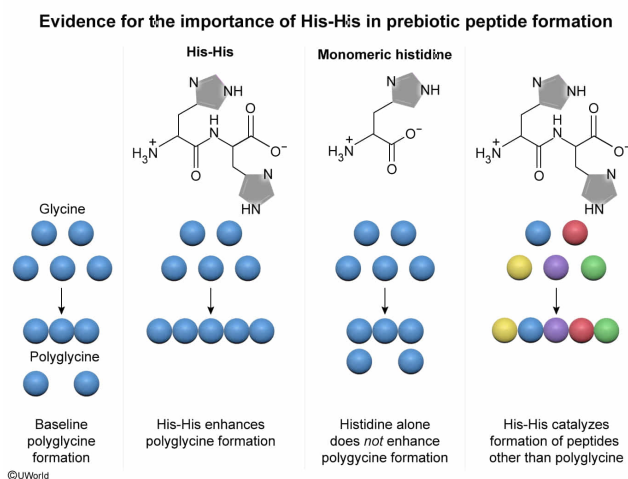
- I. A dipeptide consisting of nonhistidine amino acids increases the rate of polyglycine formation.
- II. Monomeric histidine does not increase the rate of polyglycine formation.
- III. His-His increases the rate of formation of peptides other than polyglycine.

- ☐ A. I and II only [2%]
- ☐ B. I and III only [7%]
- ✓ ☒ C. II and III only [88%]
- ☐ D. I, II, and III [2%]

Correct

87% answered correctly

Explanation:



The results of a **single experiment** may be consistent with **multiple hypotheses**. Consequently, **multiple experimental observations** are commonly used to provide additional **support for a particular hypothesis** and/or to **rule out other hypotheses**.

In this scenario, the hypothesis is that the dipeptide His-His was critical for the formation of more complex peptides on prebiotic Earth. Although this hypothesis can never be definitively proven correct, multiple observations may give support to it as a plausible hypothesis. The passage states that glycine becomes polyglycine at a faster rate when His-His is present than when His-His is absent, indicating that His-His can act as a catalyst for this reaction. However, this information *alone* does not necessarily mean that His-His was *critical* for complex peptide formation.

For instance, it is possible that monomeric histidine (ie, individual histidine amino acids) can also catalyze polyglycine formation, in which case His-His would not be necessary. Therefore, an experiment that observes that **monomeric histidine does not increase the rate of polyglycine formation** would **support the hypothesis** that His-His specifically is needed (**Number II**).

It is also possible that His-His catalyzes polyglycine formation but not the formation of other peptides. If this were the case, more complex peptide formation would *not* have been helped by the presence of His-His. However, if it were instead observed that **His-His can catalyze formation of peptides other than polyglycine**, this would **support the hypothesis (Number III)**. Many other experiments could also be performed to demonstrate (or refute) the importance of His-His but of the given options, only numbers II and III would add support to the hypothesis.

(Number I) If a dipeptide consisting of nonhistidine amino acids could increase the rate of polyglycine formation, this would show that His-His is *not* unique in its ability to catalyze these reactions, and therefore that His-His was *not* necessarily critical for prebiotic formation of complex peptides.

Educational objective:

Multiple experiments can provide stronger support for a hypothesis (or rule out alternate hypotheses) than can be provided by a single experiment. Consequently, multiple experiments are often performed to help determine which of a set of hypotheses is most likely correct.

Time spent:0 sec
Subject:
Biochemistry

QID:403686
System:
Amino Acids and Proteins

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Initial pH	his-his recovered
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6	0.36 mg
7	0.52 mg

As the initial pH of the reaction that forms His-His increases from 5 to 7, the yield of His-His increases by a factor of approximately:

- ☐ A. 0.31 [4%]
- ☐ B. 1.4 [8%]
- ☐ C. 2.3 [17%]
- ✔ ☒ D. 3.3 [69%]

Correct

69% answered correctly

Explanation:

Calculating increase in yield under different conditions

Initial pH	his-his recovered
5	0.16 mg
6	0.36 mg
7	0.52 mg

$\frac{0.52 \text{ mg}}{0.16 \text{ mg}} = 3.3$

The his-his yield at pH 7 is 3.3 X greater than the yield at pH 5

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Table 1 shows the amounts of His-His produced when histidine incubation occurs at pH levels of 5, 6, and 7. The data show that the highest yield of His-His measured occurred at pH 7 and the lowest at pH 5. The question asks for the **factor by which the yield increases** as the pH is increased from 5 to 7. In other words, the question asks **how many times larger** the pH 7 yield is than the pH 5 yield.

To determine this number, the yield at pH 7 must be divided by the yield at pH 5:

$$\frac{0.52 \text{ mg}}{0.16 \text{ mg}} = 3.25$$

In accordance with the rules of [significant figures](#), this number is rounded to 3.3. Although the exact value may be difficult to determine quickly without a calculator, it can be estimated by recognizing that 52 is approximately 3 times greater than 16 (ie, $16 \times 3 = 48$), and similarly that 0.52 is approximately 3 times greater than 0.16. Of the given answer choices, 3.3 is the closest to this approximation, and is therefore the best answer.

(Choice A) 0.31 is the result if 0.16 is divided by 0.52; in other words, the yield of His-His at pH 5 is 0.31 times as large as the yield at pH 7. However, the question asks for the opposite relation.

(Choice B) 1.4 is the factor by which the His-His yield increases when the pH is increased from 6 to 7 instead of from 5 to 7.

(Choice C) 2.3 is the factor by which the His-His yield increases when the pH increases from 5 to 6 instead of from 5 to 7.

Educational objective:

The factor by which experimental yield increases (or decreases) when conditions change can be calculated by dividing the yield of the second condition by the

yield of the original.

Time spent:0 sec

Subject:
Biochemistry

QID:403687

System:
Amino Acids and Proteins